Project Workflow

MILESTONE 1: Model Selection and Architecture

Activity 1.1: Select AI Models

- **Gemini 1.5 Pro (via API):**
  Used for: Q&A, summarization, quiz generation, learning paths
  Benefits: Advanced reasoning, structured outputs, and cloud inference
- **LaMini-Flan-T5-783M (local):**
  Used for: Concept explanation
  Benefits: Instruction-tuned, lightweight, CPU-compatible

Folder Architecture:

EduGenie/
1. main.py # FastAPI app
2. explanation_module.py # Concept explanation logic
3. qna.py # Question answering
4. quiz_module.py # Quiz generation
5. summary_module.py # Summarization
6. learning_path.py # Learning recommendations
7. templates/index.html # HTML frontend
8. static/style.css # Styling
9. requirements.txt # Python dependencies

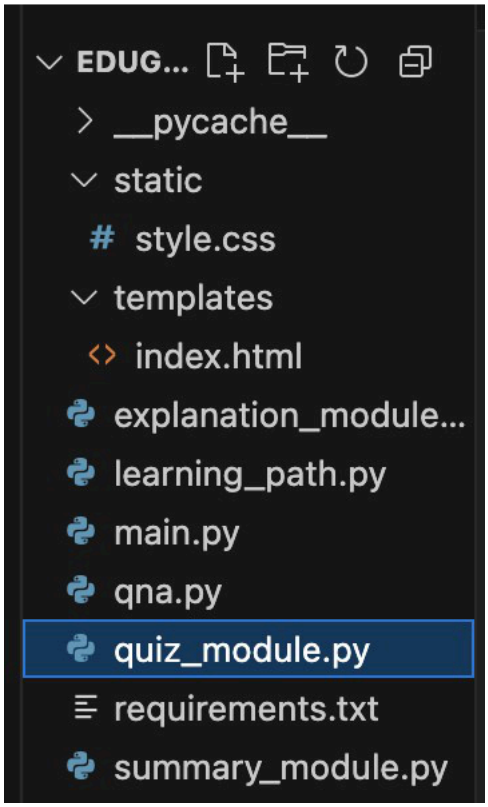


FIG. Showing the Folder Architecture

MILESTONE 2: Core Functionalities Development

Activity 2.1. : Module Implementation

1. Explanation Module

EduGenie utilizes the LaMini-Flan-T5 model, a lightweight yet powerful generative AI, to deliver educational content in a simplified and highly readable manner. This model is specifically fine-tuned to provide concise, context-aware responses that break down complex topics into easily understandable language. By focusing on clarity and brevity, EduGenie ensures that learners with minimal background knowledge or technical experience can grasp essential concepts without feeling overwhelmed. This makes it particularly valuable for beginners, school students, or self-learners seeking straightforward explanations. The integration of LaMini-Flan-T5 empowers EduGenie to act as a reliable and accessible study companion for foundational learning across subjects.